

Data Communication

Content 04: Transmission Media

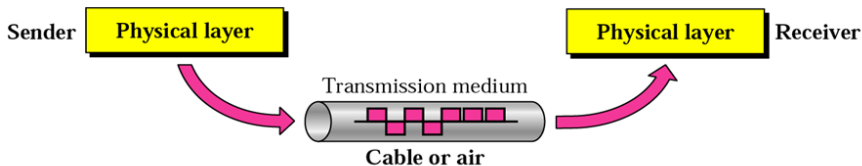
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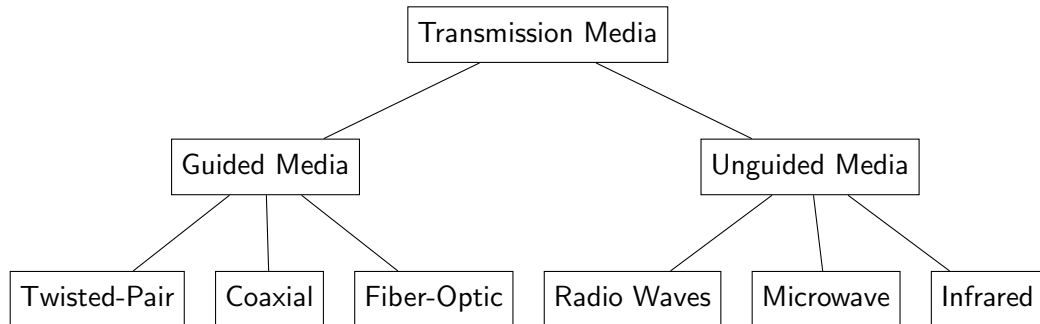


Transmission Media

Transmission media provide the physical path (or the wireless path) through which data signals travel from sender to receiver.

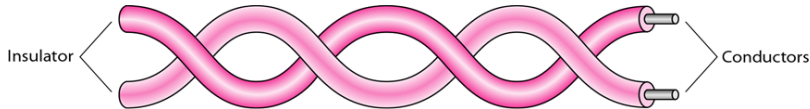


Transmission Media



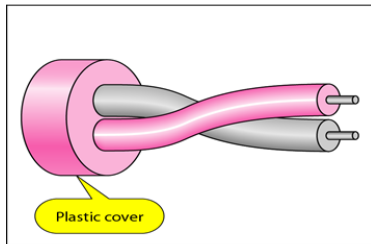
Twisted-Pair Cable

- Consists of **two insulated copper conductors** twisted together.
- One wire carries the signal, the other is the ground reference — the receiver reads the *difference* between the two.
- **Why twisted?** Twisting keeps both wires equally exposed to external noise/interference, so the induced noise cancels out when the receiver takes the difference. More twists per unit length → better noise immunity.

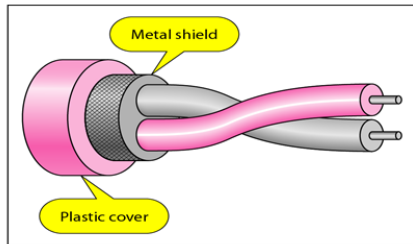


Twisted-Pair Cable: UTP vs. STP

- **UTP** (Unshielded Twisted-Pair): no extra shielding; cheaper, easier to install; most common in LANs.
- **STP** (Shielded Twisted-Pair): metallic shield around the pair reduces noise/crosstalk further; more expensive, harder to install.



a. UTP



b. STP

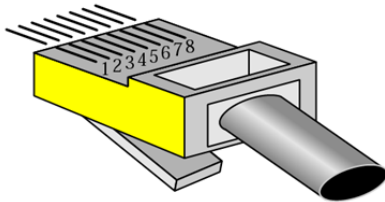
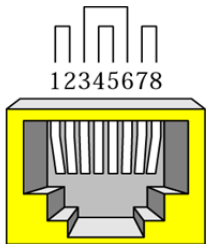
Twisted-Pair Cable: UTP Categories

Category	Data Rate	Typical Use
Cat 3	16 Mbps	Telephone lines, 10BASE-T Ethernet
Cat 4	20 Mbps	Token Ring
Cat 5	100 Mbps	100BASE-T Ethernet
Cat 5e	125 Mbps	Gigabit Ethernet
Cat 6	200 Mbps	Gigabit Ethernet
Cat 7	600 Mbps	Gigabit Ethernet / high-speed LANs



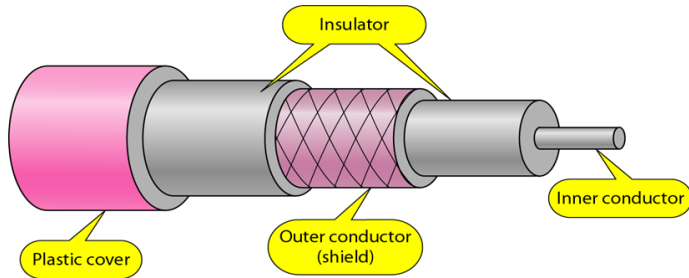
Twisted-Pair Cable: Connectors & Applications

- Most common connector: **RJ45** (Registered Jack) — a keyed connector that can only be inserted one way.
- **Applications:** Ethernet LANs (most common medium today), telephone lines, in-building structured cabling.



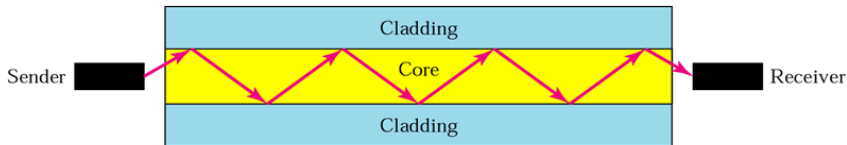
Coaxial Cable

- Carries signals over a **higher frequency range** than twisted-pair.
- Shielding gives coaxial cable better noise immunity than unshielded twisted-pair.
- **Applications:** cable television networks, legacy Ethernet, short-range broadband links.



Fiber-Optic Cable

- Made of glass or plastic; transmits data as **pulses of light** rather than electrical signals.
- **Why it works — total internal reflection:** light traveling from a denser to a less dense medium bends away from the normal as the angle of incidence increases. Beyond a *critical angle*, the light is entirely reflected back into the core instead of refracting out — this is what keeps light confined inside the fiber over long distances.



Fiber-Optic Cable: Advantages & Disadvantages

Advantages

- Noise resistance (immune to EMI)
- Less signal attenuation over distance
- Much higher bandwidth than copper

Disadvantages

- Higher cost
- Harder installation/maintenance
- Fragility



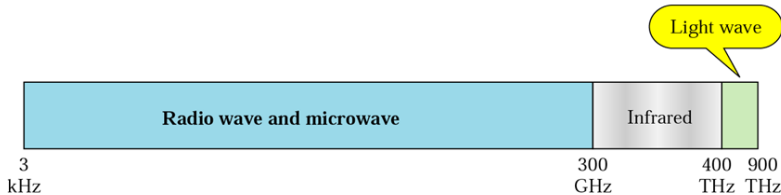
Guided Media: Summary

Medium	Bandwidth	Noise Immunity	Typical Use
Twisted-Pair	Lowest	Lowest	LAN / telephone lines
Coaxial	Medium	Medium	Cable TV / legacy Ethernet
Fiber-Optic	Highest	Highest	Backbones / long-haul links



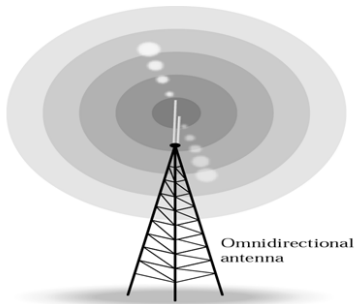
Unguided Media: The Electromagnetic Spectrum

- Unguided media transport electromagnetic waves **without a physical conductor** — this is **wireless communication**.
- Frequency ranges we focus on today: **Radio waves**, **Microwaves**, and **Infrared**.



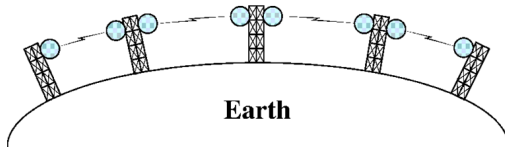
Radio Waves

- Frequency range: roughly **3 kHz to 1 GHz**.
- **Omnidirectional** — propagate in all directions from the antenna, not aimed at one specific receiver.
- Can travel long distances via sky-wave propagation.
- **Applications:** broadcast radio, television, paging systems — well suited to *multicast* communication.

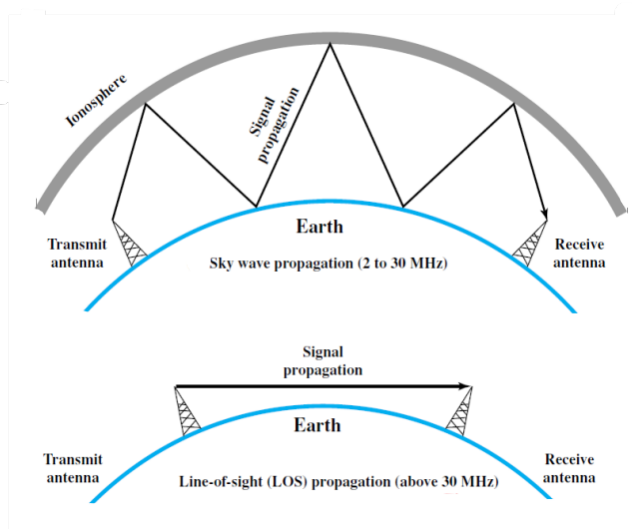


Microwaves: Terrestrial

- Frequency range: **1 GHz to 300 GHz**.
- Propagation is **line-of-sight** — antennas/towers must directly face each other.
- High-frequency microwaves cannot penetrate walls.
- The microwave band is wide (≈ 299 GHz), so it supports many sub-bands and high data rates.
- **Repeaters** extend the distance covered by terrestrial microwave links.
- **Applications:** cellular telephones, wireless LANs, point-to-point unicast links.



Sky Propagation vs Line-of-Sight Propagation

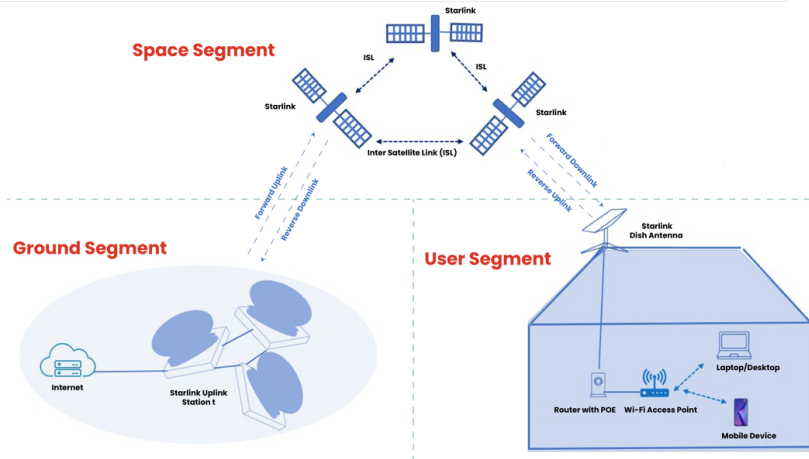


Microwaves: Satellite

- **Basic principle:** a ground station sends a signal **up** to the satellite (*uplink*); the satellite amplifies it and sends it back **down** to another location (*downlink*).
- Solves the line-of-sight distance limit of terrestrial microwave — a satellite can "see" a huge area of Earth's surface at once.
- **Applications:** satellite TV, satellite phones, satellite internet (e.g., Starlink), GPS-adjacent services, long-distance backbone links.



Satellite Communication: Starlink



Infrared

- Frequency range: **300 GHz to 400 THz**.
- Used for **short-range** communication.
- Cannot penetrate walls — this prevents interference between systems in adjacent rooms.
- Cannot be used outdoors in sunlight, since sunlight itself contains infrared that interferes with the signal.
- Requires **line-of-sight** propagation.
- **Applications:** TV/AC remote controls, short-range indoor device communication.



Unguided Media: Summary

Medium	Directionality	Penetrates Walls?	Typical Use
Radio waves	Omnidirectional	Yes	Broadcast radio/TV, paging
Microwave	Unidirectional (line-of-sight)	No	Cellular, satellite, Wi-Fi
Infrared	Unidirectional (line-of-sight)	No	Remote controls, short-range



Practice Problems

- Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
Chapter 07 - 3, 5, 8, 9, 10



References

- Data Communication and Networking (4th Ed) - Behrouz A. Forouzan
Chapter 07 - 7.1, 7.2



Thank You

